

Belcarra SPDX SBOM Overview

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Belcarra's OEM Driver Kits now include an SPDX SBOM to satisfy the NTIA Executive Order 14028.

Description

Belcarra licences a *Windows 11 NDIS/USB Filter Driver* to OEMs, allowing them to distribute the driver to their customers to use with their hardware. The driver kit is branded with the OEM's name, customized with the Vendor and Product IDs that their device(s) use.

Both kits include an SPDX SBOM manifest, in JSON format, that describes the contents of the kit. The manifest is generated using the Microsoft sbom-tool, version 4.1.5, and conforms to the SPDX 2.2 specification. The manifest is signed with Belcarra's EV Code Signing Certificate signature, and includes a SHA256 hash of the manifest file for integrity verification.

Belcarra SBOM Verifier

Belcarra makes available a browser-based SBOM verifier that can be used to verify the manifest included in the kits.

To use the Belcarra SBOM verifier open this URL: [Verifier URL](#)

The verifier will:

- unpack the kit,
- extract the manifest,
- verify the manifest's integrity using the SHA256 hash,
- verify the manifest's signature using WebCrypto and PKI.js,
- if there is a driver catalog file it will verify the package catalog signature,
- display the results along with the SPDX SBOM manifest details.

Verification Layers

The Belcarra packaging and verification flow uses four distinct verification layers:

1. `_manifest/manifest.cat` Belcarra signs the Windows catalog for the package and SBOM artifacts.
2. `manifest.spdx.json.sha256` `sbom-tool` generates the SHA-256 sidecar for `manifest.spdx.json`.
3. `manifest.spdx.json` The SBOM lists hashes for the kit contents, and the verifier recomputes those hashes for the described files outside `_manifest`.
4. Driver catalog, e.g. `belcarra.cat` In the end-user kit, the Microsoft-signed driver catalog provides the Windows driver-package signature path for the installable payload.

End-User Kit

The end-user kit, a zip file, contains the driver files, installation instructions, and license. Two bat scripts are provided:

- `setup.bat` : uses the Microsoft `pnputil` utility to install the driver on the end-user's system.
- `verify-release.bat` : uses the Microsoft `sbom-tool-x64.exe` to verify the SPDX SBOM `_manifest` included in the kit.

```
belcarra-02-05-01-001-Production-amd64_arm64-drivers.zip
├── _manifest/
│   ├── manifest.cat
│   ├── spdx_2.2/
│   │   ├── manifest.spdx.json
│   │   └── manifest.spdx.json.sha256
│   └── drivers/
│       ├── amd64/
│       │   ├── belcarrablan.pdb
│       │   ├── belcarrablan.sys
│       ├── arm64/
│       │   ├── belcarrablan.pdb
│       │   ├── belcarrablan.sys
│       ├── belcarra.cat
│       └── belcarra.inf
├── license.txt
├── README-INSTALL.txt
├── setup.bat
└── verify-release.bat
```

- [_manifest/spdx_2.2/manifest.spdx.json](#) is the SPDX SBOM for the end-user kit.
- [report-kit.json](#) is the `sbom-tool` verification report for the end-user kit.

Field	Value
SBOM Author	Belcarra Technologies (2005) Corp.
Component Name	BelcarraDemoUSBLAN
Component Version	02-05-01-001
Dependency Relationship	BelcarraDemoUSBLAN 02-05-01-001 DESCRIBES BelcarraDemoUSBLAN 02-05-01-001
Tool Name	Microsoft.SBOMTool-4.1.5
Timestamp	2026-02-13T07:43:32Z

Dependency Relationship

The end-user kit manifest describes the relationship between the SPDX document and the end-user package. When the kit is opened standalone, the verifier shows the nested package as self-described by its SPDX document.

OEM Kit

The OEM kit, a password-protected zip file, contains a README, the cfg files used to configure the release package, the end-user kit zip file, and a single bat script:

- `verify-release.bat` : uses the Microsoft `sbom-tool-x64.exe` to verify the SPDX SBOM `_manifest` included in the end-user kit zip file.

```
belcarra-02-05-01-001-Production-amd64_arm64.zip
├── _manifest/
│   ├── manifest.cat
│   ├── spdx_2.2/
│   │   ├── manifest.spdx.json
│   │   └── manifest.spdx.json.sha256
│   └── belcarra-02-05-01-001-Production-amd64_arm64-drivers.zip
├── belcarra.cfg
├── belcarra.xlsx
├── license.txt
├── README-belcarra-02-05-01-001-Production-amd64_arm64-drivers.txt
└── verify-release.bat
```

- [_manifest/spdx_2.2/manifest.spdx.json](#) is the SPDX SBOM for the OEM kit.
- [report-oem.json](#) is the `sbom-tool` verification report for the OEM kit.

Dependency Relationship

The OEM kit manifest describes the relationship between the OEM kit and the end-user kit, indicating that the OEM kit contains the end-user kit zip file as a component. The verifier carries that parent-child context into the nested view when you open the embedded end-user kit from the OEM kit report.

SBOM Minimum Elements (CISA 2025 Draft)

Field	Value
SBOM Author	Belcarra Technologies (2005) Corp.
Component Name	belcarra
Component Version	02-05-01-001
Dependency Relationship	belcarra 02-05-01-001 CONTAINS ./belcarra-02-05-01-001-Production-amd64_arm64-drivers.zip
Tool Name	Microsoft.SBOMTool-4.1.5
Timestamp	2026-02-13T07:43:37Z

Screenshots

Drag and drop the OEM kit zip file onto the webpage. You may be prompted for a password.

This verifies the manifest included in the OEM kit.

To regenerate the report screenshots and rebuild the Belcarra PDF from this repository:

```
make -C belcarra report
```

Dependency Relationship Verification

The OEM kit contains the end-user kit as a component. The verifier shows the OEM kit as the parent component, and the end-user kit as a child component. You can click on the end-user kit name to open the end-user kit zip file and verify the manifest included in the end-user kit.

Belcarra Verifier Opening and Password

Drag & Drop Package ZIP Here or Click to Select

Enter Password

This ZIP is password-protected. Enter the password to proceed.

File: belcarra-02-05-01-001-Production-amd64_arm64.zip

Show PasswordCancelOK

All processing happens locally in your browser. No files or passwords are sent anywhere.

OEM Kit Verification screenshot

Validation Passed: Processed 6 files. All file hashes match the manifest. [New File](#)

Signature & Integrity
Package: belcarra-02-05-01-001-Production-amd64_arm64.zip
_manifest/manifest.cat: Cryptographic signature verified. Signer: Belcarra Technologies (2005) Corp. (Issuer: DigiCert Trusted G4 Code Signing RSA4096 SHA384 2021 CA1, Root: DigiCert Trusted G4 Code Signing RSA4096 SHA384 2021 CA2)
SBOM Integrity: [\(info\)](#)
./manifest.spdx.json matches manifest.spdx.json.sha256
Archive Contents

```
belcarra-02-05-01-001-Production-amd64_arm64.zip
├── _manifest/
│   ├── spdx_2.2/
│   │   ├── manifest.spdx.json
│   │   ├── manifest.spdx.json.sha256
│   │   └── manifest.cat
│   ├── belcarra-02-05-01-001-Production-amd64_arm64-drivers.zip
│   ├── belcarra.cfg
│   ├── belcarra.xlsx
│   ├── license.txt
│   ├── README-belcarra-02-05-01-001-Production-amd64_arm64-drivers.txt
│   └── verify-release.bat
```

Manifest Tree [\(info\)](#) [TIMESTAMP](#) [MESSAGE](#) [EXTRA](#)

File Path	Manifest Hash (SHA256)	Status
belcarra-02-05-01-001-Production-amd64_arm64-drivers.zip	8bdc562dc59514cc...	MATCH
belcarra.cfg	4f5564673456508b...	MATCH
belcarra.xlsx	9722c3e8ff797038...	MATCH
License.txt	4260f75129bc8499...	MATCH
README-belcarra-02-05-01-001-Production-amd64_arm64-drivers.txt	7427d7f8cbb07f5c9...	MATCH
verify-release.bat	78807726d4fa5514...	MATCH

SBOM Minimum Elements (CISA 2025 Draft)

Field	Value
SBOM Author	Belcarra Technologies (2005) Corp.
Component Name	BelcarraUSBLAN-OEMKIT
Component Version	02-05-01-001
CONTAINS	belcarra-02-05-01-001-Production-amd64_arm64-drivers.zip
Tool Name	Microsoft.SBOMTool-4.1.5
Timestamp	2026-04-13T01:45:03Z

Parsed from SPDX 2.2 SBOM: createability, packages, and relationships.

Dependency Relationship Notes
*Dependency Relationship (from Parent) is contextual. It is derived when a nested kit is opened from a parent kit and is not necessarily declared inside the nested SBOM itself.
CISA (2023) Draft defines dependency relationship as inclusion or derivation between software components. SPDX reserves DESCRIBES for document-to-document relationships, so package-to-package "from parent" links are shown as CONTAINS.

End-User Kit Verification screenshot

Validation Passed: Processed 8 files. All file hashes match the manifest. [New File](#) [Back to Parent](#)

Signature & Integrity
Package: belcarra-02-05-01-001-Production-amd64_arm64.zip
Manifest: belcarra-02-05-01-001-Production-amd64_arm64-drivers.zip
_manifest/manifest.cat: Cryptographic signature verified. Signer: Belcarra Technologies (2005) Corp. (Issuer: DigiCert Trusted G4 Code Signing RSA4096 SHA384 2021 CA1, Root: DigiCert Trusted G4 Code Signing RSA4096 SHA384 2021 CA2)
SBOM Integrity: [\(info\)](#)
./manifest.spdx.json matches manifest.spdx.json.sha256
Driver Catalog Signature: [\(info\)](#)
drivers/belcarra.cat: Cryptographic signature verified. Signer: Microsoft Windows Hardware Compatibility Publisher (Issuer: Microsoft Windows Third Party Component CA 2014, Root: Microsoft Windows Third Party Component CA 2014)
Archive Contents

```
belcarra-02-05-01-001-Production-amd64_arm64-drivers.zip
├── _manifest/
│   ├── spdx_2.2/
│   │   ├── manifest.spdx.json
│   │   ├── manifest.spdx.json.sha256
│   │   └── manifest.cat
│   ├── drivers/
│   │   ├── amd64/
│   │   │   ├── belcarrablan.pdb
│   │   │   ├── belcarrablan.sys
│   │   │   ├── belcarra.cat
│   │   │   └── belcarra.inf
│   ├── license.txt
│   ├── README-INSTALL.txt
│   ├── setup.bat
│   └── verify-release.bat
```

Manifest Tree [\(info\)](#) [TIMESTAMP](#) [MESSAGE](#) [EXTRA](#)

File Path	Manifest Hash (SHA256)	Status
drivers/amd64/belcarrablan.sys	fb1515ed9e6738926...	MATCH
drivers/amd64/belcarrablan.sys	9f966cc5067f7395...	MATCH
drivers/belcarra.cat	2c588070ae761930ff...	MATCH
drivers/belcarra.inf	3a289d5256d08139...	MATCH
License.txt	4260f75129bc8499...	MATCH
README-INSTALL.txt	76bcaae227785d2...	MATCH
setup.bat	811f64acc9607a0ba...	MATCH
verify-release.bat	78807726d4fa5514...	MATCH

SBOM Minimum Elements (CISA 2025 Draft)

Field	Value
SBOM Author	Belcarra Technologies (2005) Corp.
Component Name	BelcarraUSBLAN-KIT
Component Version	02-05-01-001
CONTAINED_BY	BelcarraUSBLAN-OEMKIT 02-05-01-001
DESCRIBES	BelcarraUSBLAN-KIT 02-05-01-001
Tool Name	Microsoft.SBOMTool-4.1.5
Timestamp	2026-04-13T01:44:57Z

Parsed from SPDX 2.2 SBOM: createability, packages, and relationships.

Dependency Relationship Notes
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CISA (2023) Draft defines dependency relationship as inclusion or derivation between software components. SPDX reserves DESCRIBES for document-to-document relationships, so package-to-package "from parent" links are shown as CONTAINS.

Reports

- [report-oem.json](#) is the `sbom-tool` verification report for the OEM kit.
- [report-kit.json](#) is the `sbom-tool` verification report for the end-user kit.